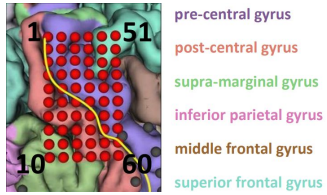
BR41N.IO

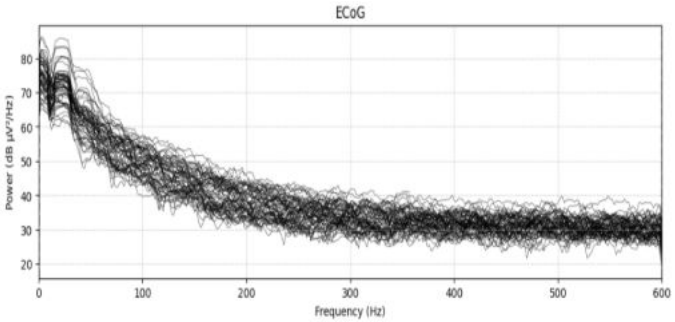
INTRODUCTION

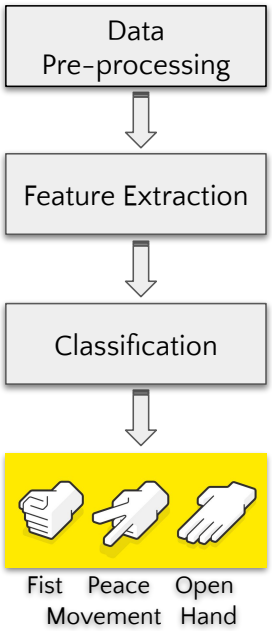
- ECoG Hand Pose Data [Sampling Frequency: 1200 Hz; 60 ECoG Electrodes; 3 Tasks (Fist, Peace, Open hand)]
- ECoG Data Dimensions [90*60*507025 -> 90*60*2400]



IDEA/SOLUTION

Power Spectral Density



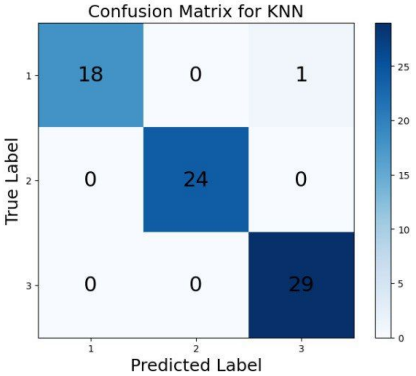


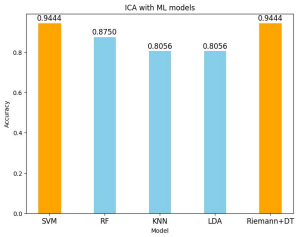
IMPLEMENTATION REALIZATION…

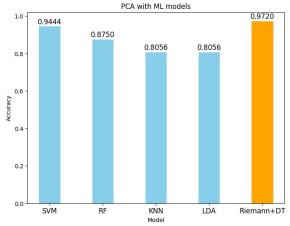
- **Data Pre-processing:** Bandpass filter (50-300 Hz), Notch filter (50 Hz), Z-score Normalization, Segmentation (0.5 secs)
- **Feature Extraction methods:** Common Spatial Patterns (CSP), Principal Component Analysis (PCA), Independent Component Analysis (FastICA), Riemann Covariances
- **Classification methods:** SVM, LDA, KNN, Random Forest, MLP, Decision Tree

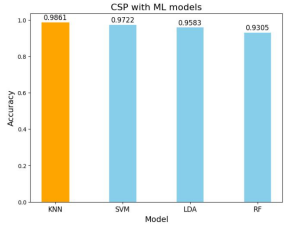
RESULTS and OUTCOME…

CSP and KNN model got highest accuracy of 98.61%, which is almost same to the result in the paper (98.9%) [Grunewald *et al.*]

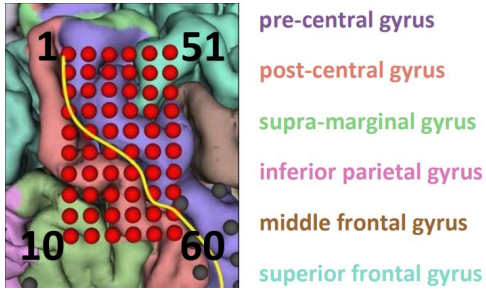


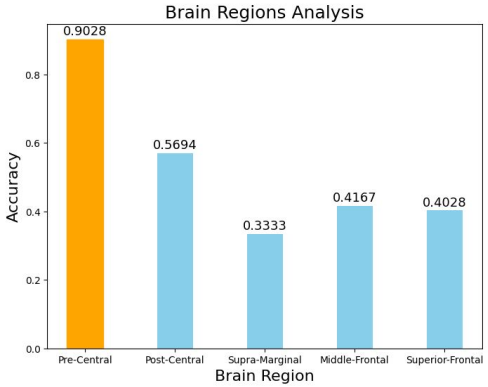


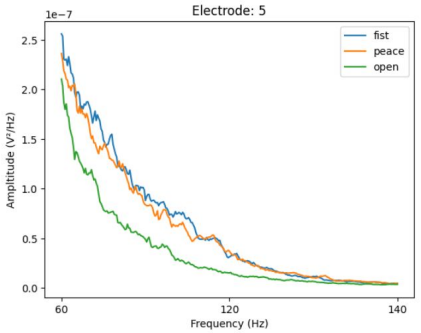


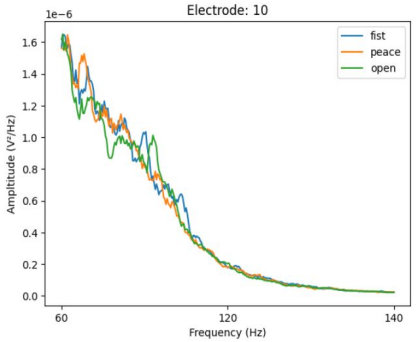


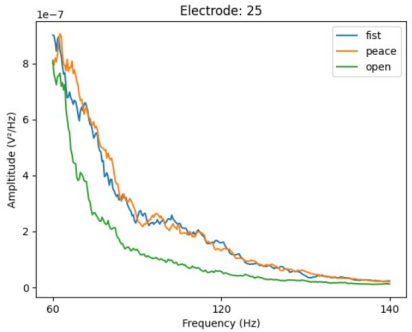
REFLECTION…

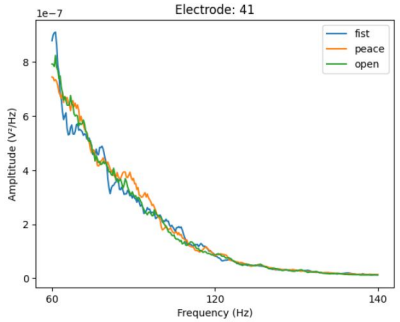

1
51
10
60
pre-central gyrus
post-central gyrus
supra-marginal gyrus
inferior parietal gyrus
middle frontal gyrus
superior frontal gyrus


Brain Regions Analysis
Accuracy
Brain Region
Pre-Central
Post-Central
Supra-Marginal
Middle-Frontal
Superior-Frontal
0.9028
0.5694
0.3333
0.4167
0.4028


Electrode: 5
1e−7
Amplitude (V²/Hz)
Frequency (Hz)
fist
peace
open


Electrode: 10
1e−6
Amplitude (V²/Hz)
Frequency (Hz)
fist
peace
open


Electrode: 25
1e−7
Amplitude (V²/Hz)
Frequency (Hz)
fist
peace
open


Electrode: 41
1e−7
Amplitude (V²/Hz)
Frequency (Hz)
fist
peace
open

Introduction

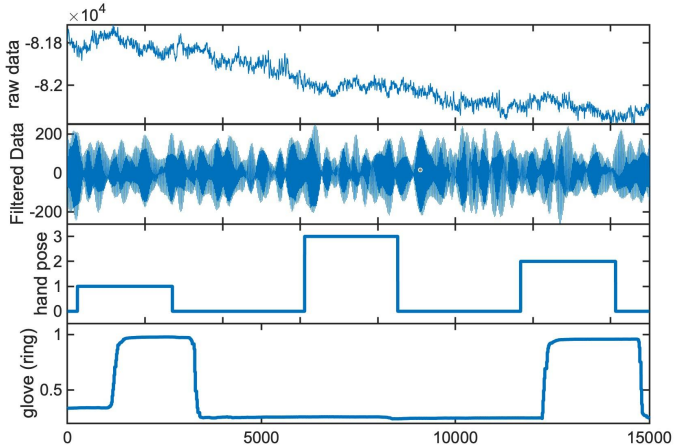
Electrocorticography (ECoG) records brain activity directly from the cortex. Hand pose classification decodes gestures from these neural signals. This project uses ECoG to classify and predict hand poses in real-time. BCIs are applied in: Medical Devices (controlling robotic arms/exoskeletons for disabled individuals); Neuroprosthetics (translating brain signals into movement for daily tasks); and Neuro Research (understanding brain motor control).

Dataset

- 90 trials of rock-paper-scissors gestures.
- ECoG: 60 Channels (C2-C61).
- Labels: 0: Rest, 1: Fist, 2: Peace, 3: Open.
- Data Glove: 5 Channels (C63-C67)
- Signal Sampling Rate: 1200 samples/sec.
- File Format: .mat.

Preprocessing

- CAR - re-referencing the data by the common average
- Notch-filter cascade up to the 6th harmonic 50 Hz
- Whitening
- Band-pass (6-th order): 50-300 Hz



Introduction

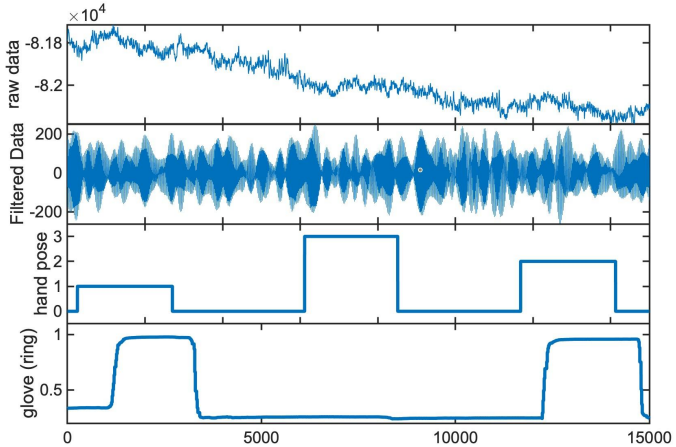
Electrocorticography (ECoG) records brain activity directly from the cortex. Hand pose classification decodes gestures from these neural signals. This project uses ECoG to classify and predict hand poses in real-time. BCIs are applied in: Medical Devices (controlling robotic arms/exoskeletons for disabled individuals); Neuroprosthetics (translating brain signals into movement for daily tasks); and Neuro Research (understanding brain motor control).

Dataset

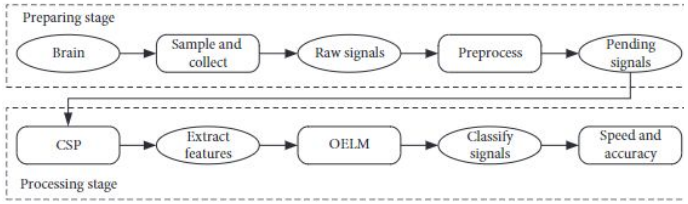
- 90 trials of rock-paper-scissors gestures.
- ECoG: 60 Channels (C2-C61).
- Labels: 0: Rest, 1: Fist, 2: Peace, 3: Open.
- Data Glove: 5 Channels (C63-C67)
- Signal Sampling Rate: 1200 samples/sec.
- File Format: .mat.

Preprocessing

- CAR - re-referencing the data by the common average
- Notch-filter cascade up to the 6th harmonic 50 Hz
- Whitening
- Band-pass (6-th order): 50-300 Hz



Literature Review

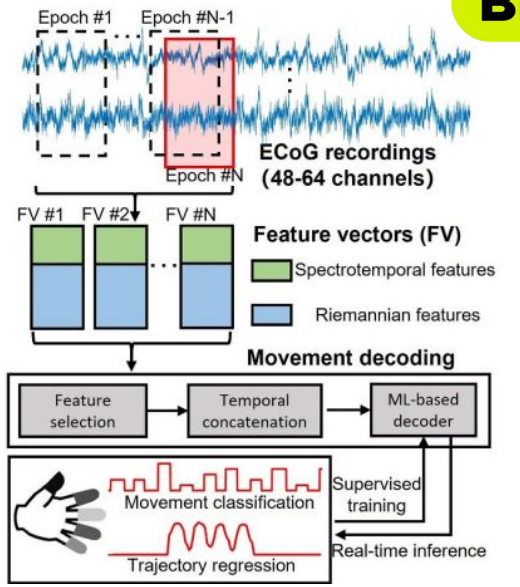


[1] An ECoG-Based Binary Classification of BCI Using OptimizedExtreme Learning Machine

[1] Zhang, X., Xiong, Q., Dai, Y., Xu, X., & Song, G. (2020). An ECoG-Based Binary Classification of BCI Using Optimized Extreme Learning Machine. Complexity, 2020(1), 2913019.]

[2] Fast and accurate decoding of finger movements from ECoG through Riemannian features and modern machine learning techniques

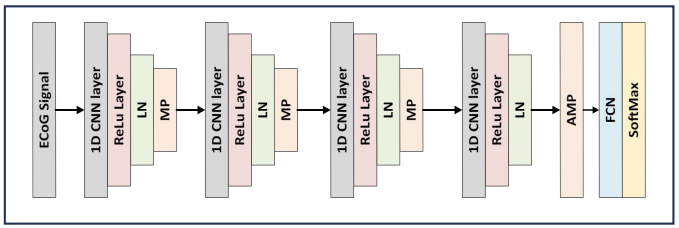
[2] Yao, L., Zhu, B., & Shoaran, M. (2022). Fast and accurate decoding of finger movements from ECoG through Riemannian features and modern machine learning techniques. Journal of Neural Engineering, 19(1), 016037.

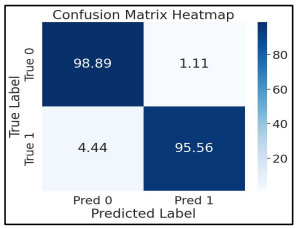


Objective

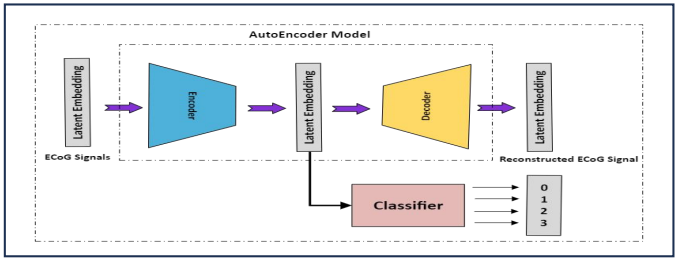
Develop a BCI system using ECoG signals for accurate, real-time hand gesture classification and trajectory prediction to improve prosthetic control and communication.

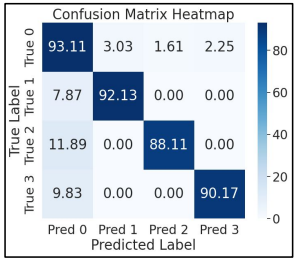
Deep Learning Model (Results)



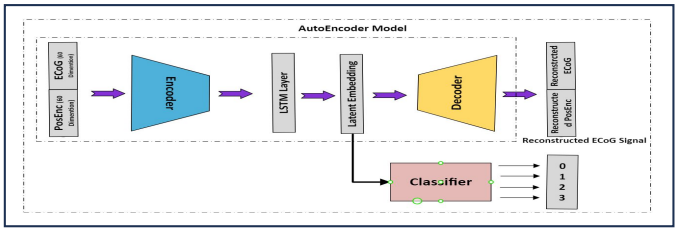


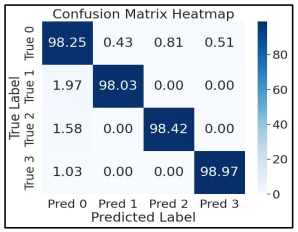
SimConv Model
97.22%
Binary (Rest/Action)





Classicoder Model
91.81%
4-class





Trajector Model
98.34%
4-class

Results

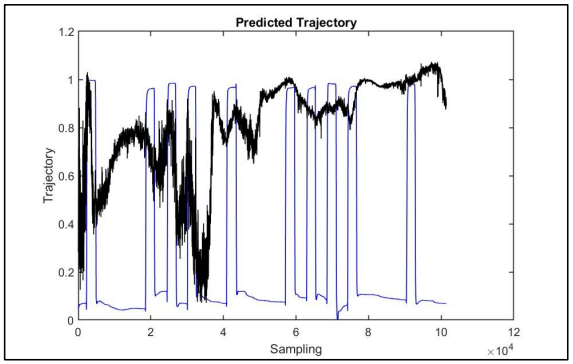


Fig: Trajectory Prediction using ECoG Signals (data glove ring)

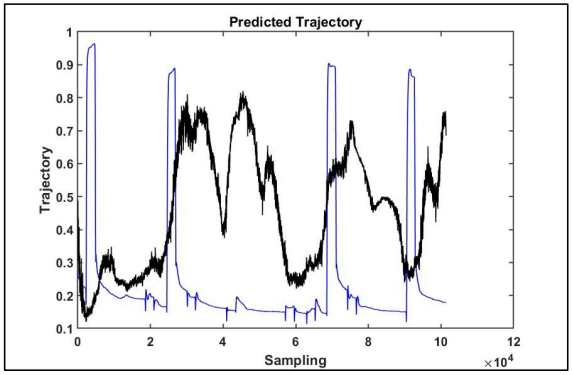
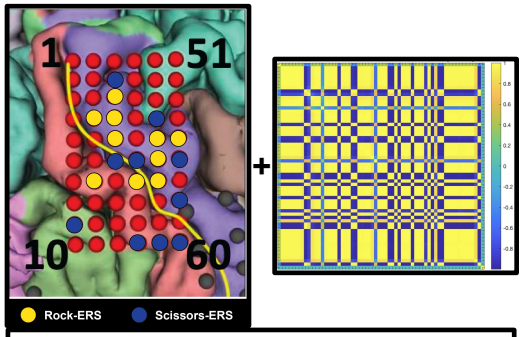


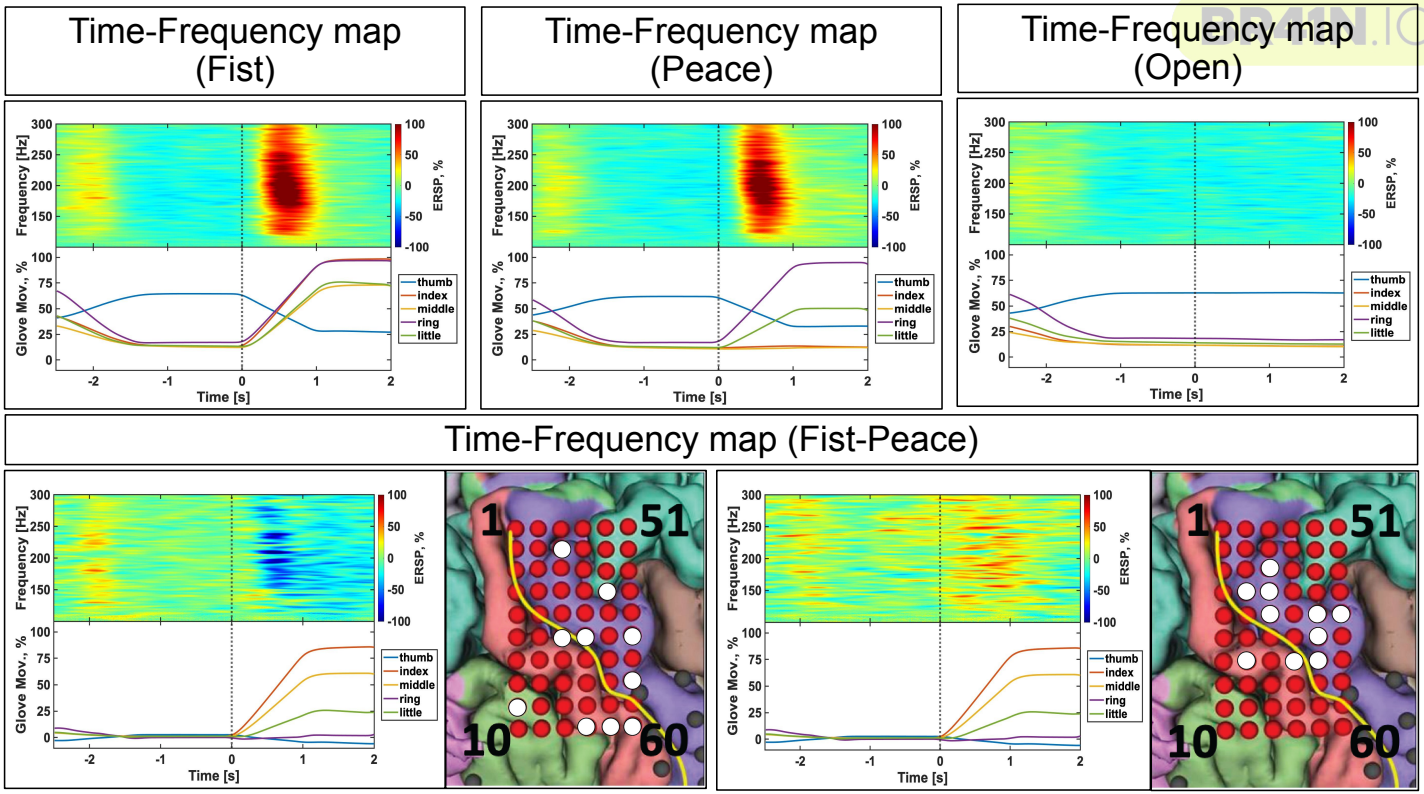
Fig: Trajectory Prediction using ECoG Signals (data glove middle)

Studies	Accuracy
IEEE SMC 2023 Hackathon	94.44 %
Summer School 2024[a]	98.61 %
IEEE SMC 2024 Hackathon[b]	98.34 %

Tab: Comparison Table with previous work



Futue Prospectus: Cross Correlation of Channels


Time-Frequency map (Fist)
Time-Frequency map (Peace)
Time-Frequency map (Open)
Time-Frequency map (Fist-Peace)
Frequency [Hz]
ERSP, %
Glove Mov., %
Time [s]
thumb
index
middle
ring
little
1
51
10
60